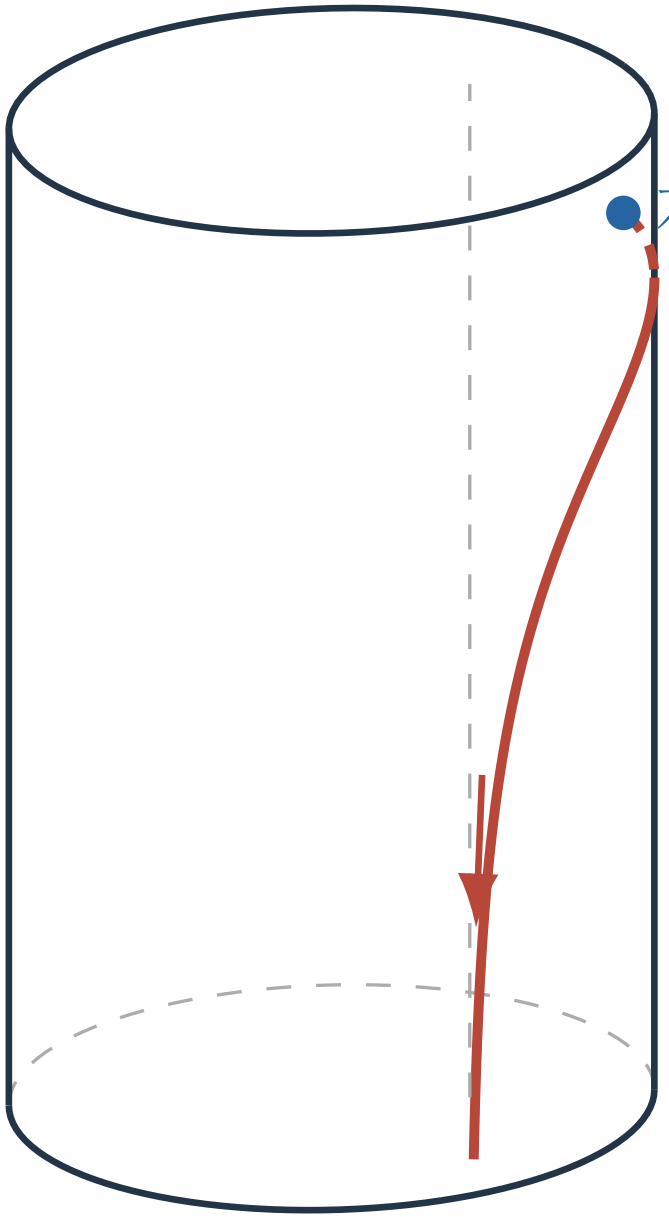
临界示例 $\omega_c = \omega_\phi = \omega > 0,\ \omega_z = 0$



$$\sin\phi = \tanh\omega t$$

$$z - z_0 = -R_0 \ln\cosh\omega t$$

$\phi \to \pi/2$

$z \to -\infty$，不闭合